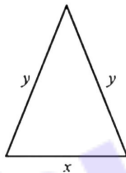


SAT UZBEKISTAN

1  Mark for Review





Note: Figure not drawn to scale.

The triangle shown has one side with a length of x inches and two sides each with a length of y inches. The perimeter of the triangle is 47 inches. Which equation represents this situation?

(A) $x + y = 47$

(B) $x + 2y = 47$

(C) $2x + y = 47$

(D) $2x + 2y = 47$

Examples

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
		.66
	.6667	0.67
	0.666	
	0.667	.67
$-\frac{1}{3}$	-1/3	
		-.33
	-.3333	
		-0.33

2 Mark for Review

If $7 + x = 4$, what is the value of $42 + 6x$?

Answer Preview:

3  Mark for Review



A batch of smoothies consists of 4 cups of milk and 2 bananas and has 1,228 milligrams (mg) of calcium. There is 304 mg of calcium in 1 cup of milk. How much calcium, in mg, is in 1 banana?

- ☐ (A) 1,216
- ☐ (B) 462
- ☐ (C) 12
- ☐ (D) 6



4 Mark for Review



A furniture company is selling all tables with a 20% discount off their original price for the month of October. What is the discount, in dollars, on a table with an original price of \$130?

- (A) 110
- (B) 90
- (C) 52
- (D) 26

5  Mark for Review

Which expression is equivalent to $(3c + 9) - (4c^3 - 5)$?

(A) $4c^3 + 3c + 14$

(B) $-4c^3 + 3c + 14$

(C) $4c^3 + 3c + 4$

(D) $-4c^3 + 3c + 4$

6

 Mark for Review

$$\begin{aligned}4x &= 16 \\ -3x + y &= -5\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

(A) -21

(B) -11

(C) 11

(D) 21

7  Mark for Review

A shopper is purchasing oranges that cost \$0.67 each. If the shopper has a coupon for \$2 off the entire purchase and there is no sales tax, which inequality represents the number of oranges, x , that the shopper can purchase for no more than \$12?

(A) $0.67x + 2 \leq 12$

(B) $0.67x + 2 \geq 12$

(C) $0.67x - 2 \leq 12$

(D) $0.67x - 2 \geq 12$

8  Mark for Review



Circle T has a radius of 35 millimeters (mm). What is the area of circle T , in mm^2 ?

(A) 35π

(B) 70π

(C) 140π

(D) $1,225\pi$

Examples

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	0.66
	.6666	.66
	.6667	0.67
	0.666	
	0.667	.67

9 Mark for Review

In the xy -plane, circle F is defined by the equation $(x + 7)^2 + (y + 7)^2 = 36$. circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 7)^2 + (y + 7)^2 = p$, where p is a constant. What is the value of p ?

Answer Preview:



10



Mark for Review



Which expression is equivalent to $4yz^3 + 5yz + 3yz^3$?

(A) $7yz^3 + 5yz$

(B) $7yz^6 + 5yz$

(C) $12yz^6$

(D) $12yz^7$

- Don't enter symbols such as a percent sign, comma, or dollar sign.



Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
	.6667	.66
	0.666	0.67

11 Mark for Review

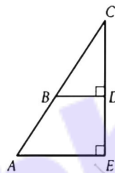
If $\frac{x}{y} = 76$ and $\frac{cx}{4y} = 76$, what is the value of c ?

Answer Preview:

12

Mark for Review

ABC



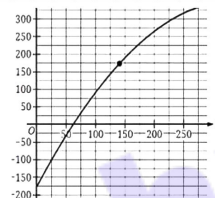
Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 21(BD)$. What is the measure of angle CAE ?

☐ (A) $(21 \cdot 57)^\circ$ ☐ (B) $(21 + 57)^\circ$ ☐ (C) 57° ☐ (D) 21°

13 Mark for Review

ABC



The graph shows the estimated boiling point y , in degrees Celsius, of a straight-chain alkane with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(141.32, 172.26)$?

- (A) A straight-chain alkane with a molecular weight of 172.26 grams per mole has an estimated boiling point of 141.32 degrees Celsius.
- (B) A straight-chain alkane with a molecular weight of 141.32 grams per mole has an estimated boiling point of 172.26 degrees Celsius.
- (C) The minimum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 141.32 grams per mole and an estimated boiling point of 172.26 degrees Celsius.
- (D) The maximum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 141.32 grams per mole and an estimated boiling point

14  Mark for Review

A landscaping company mixes oil with gasoline to power its tools. A chainsaw requires 8 ounces of oil per gallon of gasoline. A lawnmower requires 4 ounces of oil per gallon of gasoline. The company uses a total of 21 ounces of oil in its mixes for the chainsaw and lawnmower. Which of the following equations represents this situation, where x represents the amount of gasoline, in gallons, in the mix for the chainsaw and y represents the amount of gasoline, in gallons, in the mix for the lawnmower?

(A) $2(x + 2y) = 21$

(B) $2(2x + y) = 21$

(C) $4(x + 2y) = 21$

(D) $4(2x + y) = 21$



Show



Reference



More

15



Mark for Review



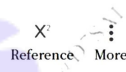
If $\frac{72}{x} = 6x$, what is the value of $6x^2$?

(A) 432

(B) 72

(C) 12

(D) 6



16

Mark for Review



$$y \leq 5x - 21$$

Which point (x, y) is a solution to the given inequality in the xy -plane?

(A) $(-5, -21)$

(B) $(0, 21)$

(C) $(5, -21)$

(D) $(6, 21)$

17  Mark for Review

A bin contains a mixture of T-shirts for two sports teams. The table shows the number of T-shirts in the bin, classified by size and sports team.

	Lions	Stars	Total
Small	6	8	14
Medium	24	34	58
Large	18	26	44
Total	48	68	116

One T-shirt from the bin will be selected at random. What is the probability of selecting a T-shirt that is medium, given that the T-shirt is a Lions T-shirt?

(A) 0.21

(B) 0.24

(C) 0.41

(D) 0.50

18  Mark for Review

$$f = \frac{(10x)^2}{7g}$$

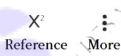
The given equation relates the positive numbers f , x , and g . Which equation correctly expresses x in terms of f and g ?

(A) $x = \frac{\sqrt{7fg}}{10}$

(B) $x = 10\sqrt{7fg}$

(C) $x = \sqrt{\frac{7fg}{10}}$

(D) $x = \sqrt{7fg} - 10$



- Don't enter symbols such as a percent sign, comma, or dollar sign.



Examples

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
	0.666	0.67

19

Mark for Review

In the xy -plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 4$. If line ℓ also passes through the point $(6, d)$, what is the value of d ?

Answer Preview:

20  Mark for Review

A circle in the xy -plane has its center at $(14, 19)$ and has a radius of $7k$. Which equation represents this circle?

(A) $(x - 14)^2 + (y - 19)^2 = 49k$

(B) $(x - 14)^2 + (y - 19)^2 = 49k^2$

(C) $(x - 14)^2 + (y - 19)^2 = 7k$

(D) $(x - 14)^2 + (y - 19)^2 = 7k^2$

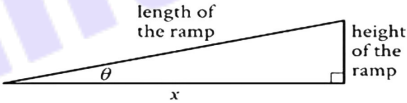
- Don't enter symbols such as a percent sign, comma, or dollar sign.



Examples

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
		.66
	.6667	
	0.666	0.67

21 Mark for Review



Note: Figure not drawn to scale.

According to a US law, ramps for use by the general public must form an angle with level ground such that $\tan\theta \leq \frac{1}{12}$. If the ramp in the figure conforms to this law and has a height of 29.1 inches, what is the least possible value of x , in inches?

Answer Preview:

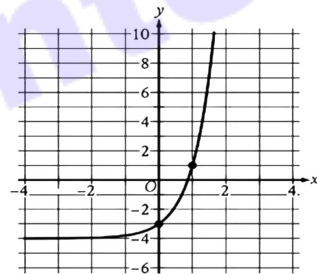
- Don't enter symbols such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
	0.666	0.67
	0.667	.67

22

Mark for Review



The equation of the graph shown is $y = a^x + b$, where a and b are constants. What is the value of $a - b$?

Answer Preview:



1 Mark for Review



$$f(x) = \frac{x + 9}{3}$$

For the function f defined by the equation shown, what is the value of $f(-6)$?

(A) -3

(B) -1

(C) 1

(D) 3

2



Mark for Review

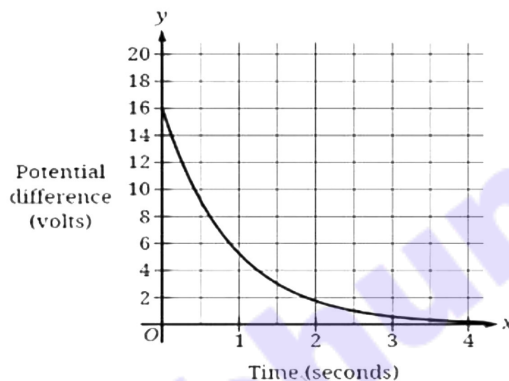


28, 28, 29, 30, 32, 33, 37

What is the mean of the data set shown?

☐ (A) 28☐ (B) 30☐ (C) 31☐ (D) 37

3 Mark for Review



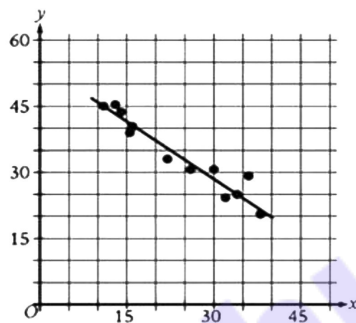
The potential difference across a capacitor in an electric circuit is initially 16 volts and decreases with time, as shown in the graph. Which of the following is closest to the time, in seconds, from the initial measurement at which the potential difference across the capacitor is 15% of its initial value?

(A) 0.43

(B) 0.85

(C) 1.71

(D) 3.41



The scatterplot shows the relationship between two variables, x and y , for data set A. A line of best fit for the data is also shown. Data set B is created by subtracting 9 units from the value of y for each data point from data set A. Which of the following is closest to the y -coordinate of the y -intercept of the line of best fit for data set B?

(A) 45.57

(B) 53

(C) 59.62

(D) 62.75

5



Mark for Review



In the linear function f , $f(0) = 5$ and $f(7) = 5$. Which equation defines f ?

(A) $f(x) = 0$

(B) $f(x) = 7$

(C) $f(x) = 5$

(D) $f(x) = x + 5$



6 Mark for Review



x	$f(x)$
1	-20
5	-40
10	-65

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- ☐ (A) $f(x) = -5(5x - 3)$
- ☐ (B) $f(x) = -5(5x - 4)$
- ☐ (C) $f(x) = -5(x + 3)$
- ☐ (D) $f(x) = -5(x - 4)$

7  Mark for Review

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 128(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 128 -gram sample starts to decay. Which statement is the best interpretation of $s(58) = 32$ in this context?

- (A) Approximately 32 grams of the sample remains 58 years after the sample starts to decay.
- (B) The mass of the sample has decreased by approximately 32 grams 58 years after the sample starts to decay.
- (C) The mass of the sample has decreased by approximately 58 grams 32 years after the sample starts to decay.
- (D) Approximately 58 grams of the sample remains 32 years after the sample starts to decay.



The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	34%
18-40 years old	23%
41-65 years old	22%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

(A) 0.21

(B) 0.32

(C) 0.45

(D) 0.62

Examples



Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	$\frac{2}{3}$.6666 .6667 0.666	0.66 .66 0.67 .67

9 Mark for Review

A right circular cone has a base diameter of 28 inches and a height of 6 inches. The volume of this cone is $k\pi$ cubic inches. What is the value of k ?

Answer Preview:



Show



Reference



More

10



Mark for Review



In triangle ABC , point D is the midpoint of segment AB and point E is the midpoint of segment BC . If $BD = 8$ and $DE = 14$, what is the length of segment AC ?

(A) 28

(B) 26

(C) 22

(D) 16

11



Mark for Review



$$x(r - 7) + 3 = 19x + 25$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

(A) 3

(B) 7

(C) 22

(D) 26

12

 Mark for Review

In a collection of items, 10% are red, 25% are green, 35% are blue, and 30% are yellow. If there are 40 green items, how many blue items are in the collection?

(A) 14

(B) 16

(C) 48

(D) 56

13



Mark for Review



$$x(13x + 27) + c = 5$$

In the given equation, c is a constant. A solution to the equation is $\frac{-27 + \sqrt{261}}{26}$. What is the value of c ?

(A) 14

(B) 9

(C) 5

(D) 4

Show

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
	.6667	.66
	0.666	0.67
		.67



14

Mark for Review

One of the factors of $2x^3 + 44x^2 + 234x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Answer Preview:

15

🔖 Mark for Review

ABC

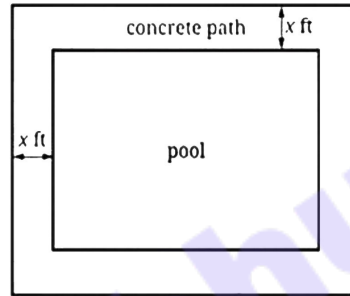
A model estimates that at the end of each year from 2015 to 2020, the number of rabbits in a population was 170% more than the number of rabbits in the population at the end of the previous year. The model estimates that at the end of 2016, there were 216 rabbits in the population. Which of the following equations represents this model, where n is the estimated number of rabbits in the population t years after the end of 2015 and $t \leq 5$?

(A) $n = 80(1.7)^t$

(B) $n = 80(2.7)^t$

(C) $n = 216(1.7)^t$

(D) $n = 216(2.7)^t$



Note: Figure not drawn to scale.

The figure shows a rectangular pool surrounded by a concrete path that is x feet (ft) wide on all sides. The pool is 21 ft long and 13 ft wide. The area of the concrete path is 240 ft^2 . What is the value of x ?

(A) 3

(B) 6

(C) 20

(D) 40

17

Mark for Review

ABC

x	y
$-2s$	17
$-s$	14
s	8

The table shows three values of x and their corresponding values of y , where s is a constant. There is a linear relationship between x and y . Which of the following equations represents this relationship?

(A) $sx + 3y = 11s$

(B) $3x + sy = 11s$

(C) $3x + sy = 11$

(D) $sx + 3y = 11$

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
	.6667	.66
	0.666	0.67
		.67
	0.667	

18 Mark for Review

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

Answer Preview:

19



Mark for Review



A square map has a side length of 30 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 65% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

(A) 5.95

(B) 10.30

(C) 19.50

(D) 48.57

20



Mark for Review



The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

I. $f(x) = a(2.1)^{x+b}$

II. $g(x) = a(2.1)^x + b$

☐ (A) I only☐ (B) II only☐ (C) I and II☐ (D) Neither I nor II



Examples



Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
	.6667	.66
		0.67

21 Mark for Review

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m ?

Answer Preview:



Examples



Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
		.66
	.6667	0.67

22

Mark for Review

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

Answer Preview: